Do Short-Term Unconditional Cash Transfers Reduce Reliance on External Borrowing? Evidence from Indonesia

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1 Introduction and Overview

Governments in developing countries frequently implement unconditional cash transfers (UCTs) to cushion households from adverse economic shocks. Whether these programs alter recipient behaviour—by changing labour supply, consumption patterns, or financial coping strategies—is a central policy question, since unintended behavioural effects could undermine the programs' welfare goals.

Al Izzati, Suryadarma, and Suryahadi (2023) examine two large-scale short-term UCTs in Indonesia: *Bantuan Langsung Tunai* (BLT), introduced in 2005 to compensate for fuel subsidy reductions, and *Bantuan Langsung Sementara Masyarakat* (BLSM), implemented in 2013–14 under similar circumstances. Using the Indonesia Family Life Survey (IFLS) and a combination of coarsened exact matching (CEM) and difference-in-differences (DiD), they find no evidence that either program significantly altered recipients' behaviour or preferences across a broad range of outcomes including employment, smoking, *arisan* participation, risk aversion, and intertemporal choice.

This paper replicates and extends Al Izzati, Suryadarma, and Suryahadi (2023). The replication reconstructs the paper's main results using IFLS waves 3, 4, and 5 (corresponding to survey years 2000, 2007, and 2014), applying the authors' CEM–DiD identification strategy with an additional methodological refinement: an expenditure-based poverty measure built from Statistics Indonesia (BPS) poverty lines. The replication confirms the original null result: 19 of 21 behavioural coefficients, and all four preference coefficients, match Al Izzati, Suryadarma, and Suryahadi (2023) in direction and statistical significance.

The extension addresses a gap the replication paper leaves open: whether short-term UCTs altered households' reliance on external borrowing—defined as loans from sources other than

*UID: u7894746, muhammad.rizki@anu.edu.au. A replication and extension of Al Izzati, Suryadarma, and Suryahadi (2023). All data are drawn from the Indonesia Family Life Survey (IFLS) and Statistics Indonesia (BPS); A special thanks to Abrar Aulia (u7808951) for helping me with the poor variable construction and getting in touch with the original Authors for the STATA code.

family or friends. A household that receives a cash transfer may reduce its borrowing needs without showing any change in employment or stated preferences. I use the IFLS' Book 2 borrowing module (Section BH) to test this question directly for the BLSM period, since IFLS4 (2007) provides a pre-BLSM borrowing baseline and IFLS5 (2014) captures post-BLSM outcomes.[1]

The results of the extension are more nuanced. Contrary to the hypothesis that transfers substitute for credit, BLSM increased external borrowing participation: receipt raised the probability of attempting to borrow by 4.5 percentage points ($p = 0.018$) and obtaining a loan by 5.0 percentage points ($p = 0.008$). The effect appears to operate through market entry, since more households attempted to borrow while the success rate among attempters did not change. A heterogeneity analysis by baseline poverty points in the direction liquidity-substitution theory predicts—a positive effect among less-poor recipients, partly offset among the poorest—but the differential is not statistically significant and is treated as suggestive only. Short-term UCTs do not appear to generate the behavioural distortions studied by Al Izzati, Suryadarma, and Suryahadi (2023), but they may still change how households manage shocks through external credit.

2 Literature Review

2.1 Behavioural effects of cash transfers

The literature on the behavioural effects of UCTs is substantial but mixed. Haushofer and Shapiro (2016) show that a one-year unconditional transfer in Kenya raised expenditure, assets, and well-being, with effects persisting years later (Haushofer and Shapiro 2018), whereas Baird, McIntosh, and Özler (2019) find the short-term impacts of a Malawi UCT dissipated within two years. Banerjee et al. (2017) find a 14-month Mexican UCT had no effect on employment or hours, and reviews by Evans and Popova (2017) and Handa et al. (2018) conclude that fears about UCTs inducing laziness or financing temptation goods are generally unfounded. Within Indonesia, Bazzi, Sumarto, and Suryahadi (2015) find timely BLT receipt supported consumption smoothing while delayed receipt reduced per-capita expenditure by 7.5 percentage points, but do not examine debt. Al Izzati, Suryadarma, and Suryahadi (2023) examine a wide range of behavioural outcomes but, again, not borrowing.

2.2 Cash transfers and credit markets

The interaction between public transfers and private credit has received less attention than the labour supply or consumption effects of UCTs. Angelucci and De Giorgi (2009), studying Mexico's Progresa, find that consumption rose among ineligible as well as eligible households in treated villages, because eligible households made more informal loans and gifts to neighbours. Cash transfers can therefore generate positive externalities in the credit market rather than simply replacing one coping strategy with another. Their setting nonetheless differs from

[1] The relevant BH variables and the IFLS3 borrowing-module limitation are detailed in Section 3.

ours: Progresa was a conditional long-term transfer, and their outcome is non-recipients' consumption rather than recipients' own borrowing.

How recipients adjust their *own* borrowing is theoretically ambiguous. Morduch (1999) argues that poor households facing missing insurance markets use informal credit to smooth consumption; a transfer supplies the liquidity such borrowing would provide, so for insurance-motivated borrowers substitution should be strong, while for investment-motivated borrowers the transfer may instead complement credit by reducing the risk of investing. His framework therefore predicts heterogeneity along the wealth distribution: the poorest should reduce borrowing most, while better-off recipients may even increase it if the transfer improves creditworthiness. Cox and Jakubson (1995) similarly find public transfers crowd out private inter-family transfers in the US, consistent with public–private substitution—though their setting maps imperfectly onto rural Indonesia. Whether the substitution logic extends to external non-family/friend borrowing under Indonesia's short-lived UCTs is an open question.

2.3 Transfers, financial inclusion, and creditworthiness

A more recent strand of work suggests the opposite of substitution may also occur: a transfer can *raise* borrowing by improving a household's access to, or standing in, the credit market, through two channels.

The first is financial inclusion. Where transfers are disbursed through formal institutions, the disbursement itself can bring excluded households into the financial system and lower the cost of subsequent borrowing. Hoffmann et al. (2024) show that switching benefit disbursement from cash to bank direct deposit increased account ownership and, notably, demand for formal loans, especially among those without a prior financial history; Bastagli et al. (2016) similarly find cash transfers are often associated with greater use of formal financial services.

The second is a creditworthiness or collateral channel: Torkelson (2020) shows that a predictable, government-verified transfer can itself serve as quasi-collateral that leads lenders to extend more credit, and Bartik et al. (2024) find a US guaranteed income raised recipients' credit scores and debt. Both channels imply that, for households not borrowing purely for survival, a transfer can be associated with *more* borrowing.

2.4 Contribution of this paper

This paper contributes to the literature in three ways. First, it provides the first direct test of whether Indonesia's short-term UCTs altered household external borrowing behaviour, using a CEM–DiD design that mirrors the identification strategy of Al Izzati, Suryadarma, and Suryahadi (2023). The borrowing outcome is a natural complement to the employment and preference outcomes the original paper examines: a program that does not change behaviour in the labour market or in preferences may nonetheless alter how households manage their short-run financial position, with potentially important implications for debt accumulation and financial vulnerability.

Second, the paper explores whether the borrowing effect varies by baseline poverty status, motivated by Morduch's (1999) prediction that liquidity effects should be strongest for the

most financially constrained households.

Third, the replication itself contributes by implementing several methodological refinements over a straightforward reconstruction of Al Izzati, Suryadarma, and Suryahadi (2023): an expenditure-based poverty measure using BPS poverty lines rather than the self-reported poor card, the BPS urban/rural classification taken from community tracking records, and exact education category codes in the matching specification following the authors' original program code.

3 Data and Methods

3.1 Data

The analysis uses three waves of the Indonesia Family Life Survey (IFLS), a publicly available household panel representative of approximately 83% of the Indonesian population (Strauss, Witoelar, and Sikoki 2016). The replication follows Al Izzati, Suryadarma, and Suryahadi (2023) and uses two program periods. Dataset 1 studies BLT I, with IFLS3 (2000) as the pre-program wave and IFLS4 (2007) as the post-program wave. Dataset 2 studies BLSM, with IFLS4 (2007) as the pre-program wave and IFLS5 (2014) as the post-program wave.

Treatment is identified from the household social-assistance module. Households with a BLT or BLSM card (KR27B) are coded as recipients. The timing of both programs was driven by national fuel-subsidy reforms and is plausibly exogenous to any individual household; household receipt, however, was targeted towards poorer households and is therefore not random. The empirical strategy therefore relies on matching and fixed effects, rather than on random assignment.

For the replication, outcomes are measured at the individual level and follow Al Izzati, Suryadarma, and Suryahadi (2023): employment, smoking, *arisan* participation, working hours, side jobs, risk aversion, and intertemporal choice are drawn from IFLS Books 3A and 3B. For the extension, outcomes are measured at the household level because the IFLS borrowing module is recorded at the household level. Household borrowing outcomes are drawn from Book 2, Section BH, and are summarised in Table 1.

Three data limitations follow. First, IFLS3 (2000) does not contain the BH borrowing module, so the extension can only be estimated as a BLSM-period DiD using IFLS4–IFLS5; the BLT I borrowing analysis is therefore limited to a matched IFLS4 cross-section and is reported as descriptive evidence only. Second, because the borrowing module is household-level, the extension is estimated at the household level and CEM is implemented using household-level baseline characteristics. Third, repayment and outstanding-balance are not fully comparable across waves: IFLS4 records the largest loan, while IFLS5 records all loans. These intensive-margin outcomes are therefore interpreted cautiously.

3.2 Poverty status construction

A key matching variable in Al Izzati, Suryadarma, and Suryahadi (2023) is household poverty status, but the original paper does not fully document how this variable is constructed, and the

Table 1: Extension outcome variables from IFLS Book 2, Section BH

Outcome	IFLS variable(s)	Definition
Attempted borrowing	BH02	Equals 1 if any household member attempted to borrow from a non-family/friend source in the past 12 months.
Obtained loan	BH07	Equals 1 if the household obtained a loan; non-attempters are coded as 0.
Denied loan	BH04	Equals 1 if the household was turned down; non-attempters are coded as 0.
Number of episodes	BH09	Number of external borrowing episodes, coded as 0 for non-borrowers.
Amount borrowed	BH10	Total amount borrowed, transformed as $\log(x+1)$ in the regressions.
Amount repaid	BH22/BH27	Amount repaid, transformed as $\log(x+1)$.
Outstanding balance	BH24/BH28	Outstanding balance, transformed as $\log(x+1)$.

Notes: BH22 and BH24 are used in IFLS4; BH27 and BH28 are used in IFLS5. Borrowing refers to non-family/friend sources. 'Obtained loan" and 'Denied loan" are defined unconditionally over the matched household panel by coding non-attempters as zero. Conditional success and denial rates among attempters are reported separately as secondary descriptive checks.

IFLS does not provide a ready-made expenditure-poverty indicator suitable for this purpose. I therefore construct an expenditure-based poverty indicator for IFLS4 and IFLS5 using official BPS poverty-line publications: Badan Pusat Statistik (2008) for 2007 and Badan Pusat Statistik (2014) for 2014. These publications report monthly per-capita poverty lines by province, urban/rural status, and year.

I first construct monthly household expenditure from the IFLS expenditure modules. Total expenditure is the sum of weekly food spending converted to a monthly value, monthly non-food spending, annual non-food spending divided by 12, annual school spending divided by 12, and monthly housing expenditure. I then merge each household to the relevant BPS poverty line using province, urban/rural status, and survey year. Because the BPS line is per capita, I convert it into a household-specific poverty threshold by multiplying it by household size. A household is classified as poor if this threshold exceeds total monthly household expenditure:

$$Poor_h = 1\{PL_{p,u,t} \times HHSize_h > HHExp_h\},$$

where $PL_{p,u,t}$ is the BPS poverty line for province p, urban/rural status u, and year t; $HHSize_h$ is roster-based household size; and $HHExp_h$ is total monthly household expenditure. Households without a valid household-size count are left unclassified rather than coded poor.

The resulting expenditure-poor share is small: about 2.1% of households in 2007 and 1.7% in 2014 in the full merged files. Because roughly 30% of households lack a valid household-size count, the share among households with non-missing poverty status is slightly higher, but still small. This strict BPS-based measure is useful for identifying the poorest households, but it also means that the poverty heterogeneity analysis is underpowered.

The constructed poverty indicator, `poor`, is used as a CEM matching covariate in both the replication and the extension, alongside the self-reported poor-card indicator. In the extension,

I also use its baseline value to define the heterogeneity term in equation 2.

3.3 Coarsened exact matching

The main identification problem is that BLT and BLSM were targeted towards poorer households. A simple comparison of recipients and non-recipients would therefore confound program receipt with baseline disadvantage. Following Al Izzati, Suryadarma, and Suryahadi (2023), I use coarsened exact matching (CEM) to improve comparability between treated and control observations before estimating DiD models.

For the individual-level replication, CEM is implemented at baseline for each program period. The matching covariates are sex, education, house ownership, safe water, septic toilet, farmland, urban residence, the constructed expenditure-poverty indicator, the self-reported poor-card indicator (*surat miskin*), crisis-assistance receipt for Dataset 1, and province. The CEM procedure produces 1,150 strata for Dataset 1, of which 506 are matched, and 1,860 strata for Dataset 2, of which 820 are matched.

For the borrowing extension, CEM is implemented at the household level because the borrowing outcomes are measured at the household level. The extension CEM uses household-level baseline characteristics: sex and education of the household respondent, house ownership, safe water, septic toilet, farmland, urban residence, the constructed poverty indicator, the poor-card indicator, and province. This produces 4,777 household strata, of which 802 are matched. All regressions use the CEM weights generated by the matching procedure.

Table 6 reports baseline imbalance before CEM. The large pre-matching differences confirm that recipients were poorer, less educated, more rural, and had worse access to safe water and sanitation. After CEM, multivariate L1 imbalance falls to approximately zero in both replication datasets and in the household-level extension, indicating exact balance on the coarsened matching variables.

3.4 Model specification

For the replication, I estimate the same two-period DiD model as Al Izzati, Suryadarma, and Suryahadi (2023) within the CEM-matched sample:

$$y_{it} = \alpha + \beta(UCT_i \times Post_t) + \delta_i + \tau_t + \varepsilon_{it}, \tag{1}$$

where y_{it} is the outcome for individual i in wave t, UCT_i indicates program receipt, $Post_t$ is the post-program indicator, δ_i is an individual fixed effect, and τ_t is a period fixed effect. The coefficient of interest is β, the DiD estimate of the effect of UCT receipt. Standard errors are clustered at the individual level. For pooled behavioural estimates combining Dataset 1 and Dataset 2, I follow the original paper and include a dataset-by-post control.

For Dataset 2, some households had prior exposure to BLT I before the BLSM period. Because prior BLT I receipt is time-invariant, its level is absorbed by the fixed effect. I therefore allow prior recipients to follow a different post-period trend by including $Post_t \times PriorUCT_i$ in the relevant Dataset 2 specifications.

For the extension, the same DiD logic is applied at the household level:

$$y_{ht} = \alpha + \beta(UCT_h \times Post_t) + \mu_h + \tau_t + \varepsilon_{ht},$$

where h indexes households. The panel unit is the full household identifier, household fixed effects remove time-invariant household heterogeneity, and standard errors are clustered at the household level. Binary borrowing outcomes are estimated as linear probability models for consistency with the replication paper.

To test whether the BLSM effect differs by baseline poverty status, I augment the household-level specification with a triple interaction:

$$y_{ht} = \alpha + \beta(UCT_h \times Post_t) + \theta(UCT_h \times Post_t \times Poor_h) + \delta(Post_t \times Poor_h) + \mu_h + \tau_t + \varepsilon_{ht}, \quad (2)$$

where $Poor_h$ is the baseline expenditure-poverty indicator. Here, β is the BLSM effect for non-poor households, while θ is the additional effect for poor households. The net BLSM effect for poor households is therefore $\beta + \theta$. The term $Post_t \times Poor_h$ allows poor and non-poor households to have different post-period changes even in the absence of treatment.

4 Replication Results and Discussion of the Original Report

4.1 Pre-matching balance

Table 6 reports baseline characteristics by UCT receipt status for both datasets. Before matching, recipients are less educated, more likely to hold a poor card, more rural, and have lower access to safe water and proper sanitation than non-recipients, confirming the programs' targeting and the need for matching. After CEM all covariate differences are eliminated by construction (multivariate L1 ≈ 0). Figure 1 (Appendix) plots the Dataset 2 differences with 95% confidence intervals; only sex and junior-secondary education are statistically indistinguishable across groups before matching.

4.2 Behavioural effects (replication of Table 4)

Table 2 sets my replicated DiD estimates of BLT I and BLSM receipt side by side with the corresponding coefficients in Al Izzati, Suryadarma, and Suryahadi (2023), across all three panels (pooled, BLT I, and BLSM). The coefficient of interest throughout is $\hat{\beta}$ on the interaction $UCT_i \times Post_t$, and a "match" (✓) records agreement in statistical significance at the 5% level.

Panel B (BLT I) replicates the paper closely. *Arisan* participation is positive but insignificant (0.015; paper: 0.009), so BLT I did not crowd out informal rotating savings. Working hours among the employed are essentially zero (0.075), side jobs are zero (0.000), and the summary index (0.012) is small and insignificant, close to the paper's 0.031: the overall behavioural effect of BLT I is negligible and precisely estimated as zero. Two outcomes diverge—the cigarette-inversion proxy and the employment dummy—and I return to both in Section 4.4.

Panel C (BLSM) replicates the paper cleanly: *arisan* (0.003), smoking cessation (-0.026), cigarettes (0.002), employment (-0.004), working hours (-1.519), side jobs (0.003), and the

summary index (−0.015) are all insignificant, matching the original on every outcome. Panel A (pooled) confirms the picture: every behavioural outcome, including employment (−0.013, SE = 0.008), is insignificant, since pooling absorbs the BLT-specific employment deviation. The combined evidence matches the paper's core finding that short-term UCTs do not systematically alter behaviour.

Table 2: Replication comparison: original vs. replicated DiD estimates ($UCT \times Post$)

	Panel A: Pooled		Panel B: BLT I		Panel C: BLSM		
Outcome	Paper	Ours	Paper	Ours	Paper	Ours	Match?
Arisan	0.010	0.006	0.009	0.015	0.011	0.003	✓
Smoke quit	0.001	−0.015	0.003	0.005	−0.002	−0.026	✓
Cigs (inverted)	0.007	0.004	0.019	**0.008****	−0.012	0.002	×
Working	0.007	−0.013	0.013	**−0.024***	−0.004	−0.004	×
Hours worked	−0.600	−0.494	−1.333	0.075	0.520	−1.519	✓
Side job	0.006	−0.001	0.010	0.000	0.000	0.003	✓
Summary index	0.021	−0.016	0.031	0.012	0.006	−0.015	✓

Notes: "Paper" reproduces the $UCT \times$ year coefficients from Al Izzati, Suryadarma, and Suryahadi (2023) Table 4; "Ours" reports the replicated DiD coefficient on $UCT \times Post$. The **Match?** column reports agreement in statistical significance at the 5% level: ✓ if my estimate agrees with the paper in every panel, × if it deviates in any panel. **Bold** marks the specific deviating cell—both occur in the BLT I panel: the cigarette proxy (significant where the paper is not) and the employment dummy (negative and significant where the paper is positive and insignificant). All regressions include year and individual fixed effects with CEM weights; Panel A adds a dataset fixed effect interacted with the post indicator. Cigarettes, hours, and side jobs are estimated among smokers/workers conditional on baseline status. $^{*}p < 0.05$; $^{**}p < 0.01$.

4.3 Preferences (replication of Table 5)

Table 7 reports DiD estimates on risk aversion and intertemporal choice for Dataset 2. All four preference outcomes are insignificant and consistent with the paper's null results: risk aversion Set 1 (−0.014, SE = 0.039) and Set 2 (−0.292, SE = 1.016), and intertemporal preference Set 1 (0.035, SE = 0.037) and Set 2 (0.036, SE = 0.043).

4.4 Discussion of deviations

My replication leaves two remaining deviations, both in the BLT I panel. First, the employment estimate is negative and significant (−0.024, SE = 0.010), whereas the original paper reports a positive but insignificant effect. This likely reflects differences in matching stringency: my CEM retains about 97% of treated observations, compared with the paper's 87%, and the deviation disappears in the pooled estimate. Second, the BLT cigarette-intensity proxy is statistically significant but economically small (0.008). Overall, 19 of 21 behavioural outcomes and all four preference outcomes match the original paper in statistical significance, supporting the core null result.

5 Extension: Do Short-Term UCTs Reduce External Borrowing?

5.1 Motivation and hypotheses

Although Al Izzati, Suryadarma, and Suryahadi (2023) show that short-term UCTs did not generate labour-supply, smoking, or informal-insurance distortions, their analysis leaves open an important behavior: do transfers change households' reliance on external borrowing? A transfer may shift reliance on external credit without showing up in labour-market or preference outcomes. This matters because external borrowing is itself welfare-relevant—debt taken on to survive a shock raises later repayment burdens—and because the interaction between formal transfers and credit markets is underexplored in Indonesia. Bazzi, Sumarto, and Suryahadi (2015) show BLT supported consumption smoothing but do not ask whether it operated through reduced borrowing.

Drawing on Morduch's (1999) framework, I expect the mechanism to vary with baseline wealth. Poor households rely heavily on informal credit as a substitute for insurance: when income falls short of consumption needs, borrowing fills the gap, and a transfer arriving during or after a shock can directly substitute for this borrowing. Better-off households within the eligible population are less likely to be borrowing for survival; their borrowing is more likely to be investment-driven and may be unaffected by the transfer, or even stimulated if the transfer improves their creditworthiness, as in the financial-inclusion and collateral mechanisms documented by Hoffmann et al. (2024), Torkelson (2020), and Bartik et al. (2024) (Section 2.3).

I test two hypotheses. **Hypothesis 1**: BLSM receipt reduces the probability that a household attempts to borrow from non-family/friend sources and reduces outstanding external debt. **Hypothesis 2**: Any debt-reducing effect is stronger among poorer households, for whom the transfer represented a larger share of baseline expenditure and whose borrowing is most likely insurance-motivated rather than investment-motivated. A null result for the aggregate effect would still be informative: it would extend the Al Izzati, Suryadarma, and Suryahadi (2023) conclusion from behavioural and preference outcomes to financial coping behaviour.

5.2 Descriptive statistics

Among matched IFLS4 households, about 17% attempted external borrowing in the past 12 months at baseline and 17% obtained a loan; by 2014 the attempt rate had risen to roughly 28%, reflecting Indonesia's financial deepening over the period. This rising trend means that detecting a treatment-group *reduction* would require the UCT to offset the general increase—making support for Hypothesis 1 harder, not easier, to find within the DiD framework.

5.3 Extensive-margin results

Table 3 reports DiD estimates of BLSM receipt on extensive-margin borrowing outcomes, all defined over the full matched household panel. BLSM receipt raised the probability of attempting to borrow by 0.045 (SE $= 0.019$, $p = 0.018$) and the probability of obtaining a loan by 0.050 (SE $= 0.019$, $p = 0.008$), both *positive*—the opposite of Hypothesis 1. The probability of

being denied a loan is a precise zero (-0.007, SE $= 0.006$), and the extensive-margin summary index is positive and significant (0.127, SE $= 0.050$, $p = 0.012$).

Attempting and obtaining a loan are not two independent findings: in this sample the two are correlated at 0.98, because conditional on attempting, households almost always succeed (the within-sample success rate among attempters is about 97%). The attempt and success coefficients therefore describe a single shift on the extensive margin rather than two separate channels. What distinguishes *how* that shift occurs is the success rate *among households that attempted*, which is unchanged ($+0.023$, SE $= 0.024$, $p = 0.34$; Appendix Table 8). Taken together, the results indicate that BLSM raised successful borrowing almost entirely by bringing more households into the credit market—more households tried to borrow, and they succeeded at the same rate as before—rather than by raising approval odds among households that would have applied anyway.

These results reject Hypothesis 1 for the extensive margin. The pattern is consistent with the financial-inclusion and creditworthiness mechanisms in Section 2.3: disbursement may have introduced households to formal channels and lowered the cost of approaching lenders (Hoffmann et al. 2024; Bastagli et al. 2016), or government-verified recipient status may have signalled creditworthiness (Torkelson 2020; Bartik et al. 2024). I stress that these are *consistent with*, not demonstrated by, the data: the IFLS records neither disbursement channel nor lender type, so the analysis establishes that credit-market participation rose without identifying which mechanism produced it.

Table 3: BLSM effect on external borrowing: extensive margin

	(1) Attempted	(2) Obtained loan	(3) Denied	(4) Index
UCT × Post	0.045*	0.050**	−0.007	0.127*
	(0.019)	(0.019)	(0.006)	(0.050)
Observations	9,945	9,945	9,945	9,945

Notes: Household fixed effects, period fixed effects, CEM sample weights; standard errors clustered at the household level. All outcomes are unconditional over the matched panel: "Obtained loan" and "Denied" are coded zero for households that did not attempt to borrow. The index is a standardised average of columns 1 and 2 following Kling, Liebman, and Katz (2007). Matched sample: 802 of 4,777 strata; 3,608 control and 1,514 treated baseline households (about 67% of treated BLSM households retained). The conditional success/denial estimates (among attempters only) are reported separately in Appendix Table 8. *$p < 0.05$; **$p < 0.01$.

5.4 Intensive-margin results

Table 4 reports DiD estimates on intensive-margin outcomes, measured over the matched panel with non-borrowers coded zero and amounts in $\log(x + 1)$. The number of borrowing episodes is insignificant (-0.046, SE $= 0.105$). Log amount borrowed (0.362, SE $= 0.142$, $p = 0.011$) and log amount repaid (0.907, SE $= 0.412$, $p = 0.028$) are positive and significant, while log outstanding balance is insignificant (0.606, SE $= 0.401$). Because these outcomes

include zeros for non-borrowers, the significant amount and repayment coefficients largely reflect the same extensive-margin shift already documented—more households holding a positive (rather than zero) loan amount—rather than independent evidence that BLSM raised the scale of borrowing among those who would have borrowed regardless. Read this way, the intensive margin reinforces the participation story rather than adding a separate one.

Table 4: BLSM effect on external borrowing: intensive margin

	(1) # Episodes	(2) Log amount	(3) Log repaid	(4) Log outstanding
UCT × Post	−0.046	0.362*	0.907*	0.606
	(0.105)	(0.142)	(0.412)	(0.401)
Observations	9,945	9,945	9,945	9,945
Sample	All	All	All	All

Notes: Household fixed effects, period fixed effects, CEM weights. All outcomes are measured over the matched panel with non-borrowers coded zero; log transformations use $\log(x+1)$, so the significant amount and repaid coefficients largely reflect the extensive-margin shift documented in Table 3 rather than a change in scale among continuing borrowers. Column 3 (repaid) is additionally not directly comparable across waves: IFLS4 records repayment on the largest loan only; IFLS5 records the aggregate across all loans. *$p < 0.05$; **$p < 0.01$.

The repayment result additionally reflects a measurement-scope change noted in Section 3: IFLS4 records repayment on the largest loan only, while IFLS5 records the total across all loans, so part of the increase may be artefactual.

5.5 Heterogeneity by baseline welfare

Table 5 reports the heterogeneity estimates from equation (2). For borrow attempt, the $UCT \times Post$ coefficient among less-poor households (those above the expenditure-poverty threshold) is positive and significant (0.062, SE $= 0.021$, $p = 0.003$); the corresponding estimate for obtaining a loan is also significant (0.067, SE $= 0.021$, $p = 0.001$). The triple interaction $UCT \times Post \times Poor$ is negative for both outcomes (-0.141 for attempt, -0.146 for success), in the direction liquidity-substitution theory predicts, but neither is statistically significant ($p = 0.28$ and $p = 0.26$). The net effect for expenditure-poor households, recovered by a linear combination $\beta + \theta$, is -0.079 (SE $= 0.129$, $p = 0.54$) for attempt—negative in sign but very imprecisely estimated and far from significant.

I therefore treat the welfare heterogeneity as *suggestive only*. The point estimates are consistent with Morduch's (1999) prediction—a positive credit-participation effect among less-poor recipients, partly offset among the poorest, for whom a transfer more plausibly substitutes for survival borrowing—but the data cannot reject the null of a common effect across the welfare distribution. The imprecision has a clear source: under the corrected expenditure-poverty measure only about 2% of matched households are classified poor, so the poor cell is small and the interaction is weakly identified. The honest conclusion is that the robust finding of this paper is the *average* extensive-margin increase in credit-market participation; whether that

increase differs by baseline poverty is a question this sample is underpowered to settle.

Table 5: Heterogeneity by baseline welfare (suggestive)

	(1) Attempted	(2) Obtained loan	(3) Log outstanding
UCT × Post	0.062**	0.067**	0.407
	(0.021)	(0.021)	(0.431)
UCT × Post × Poor	−0.141	−0.146	2.419
	(0.131)	(0.131)	(1.620)
Net effect for poor ($\beta + \theta$)	−0.079	−0.079	2.826
(SE; p)	(0.129; 0.54)	(0.129; 0.54)	(1.561; 0.07)
Observations	8,454	4,752	5,701

Notes: *Poor* is the expenditure-based indicator (BPS poverty line × household size > monthly expenditure in 2007) and is a CEM matching covariate in the extension; in this specification its baseline value additionally enters as the triple-interaction moderator. The specification includes *Post* × *Poor*. The net poor effect $\beta + \theta$ and its SE/p are from a post-estimation linear combination (`lincom`). None of the triple interactions or net poor effects is significant at conventional levels; the table is reported as suggestive evidence only (see text). Household and period fixed effects; CEM weights; SEs clustered at the household level. *$p < 0.05$; **$p < 0.01$.

5.6 Sensitivity and interpretation

The main extensive-margin result is defined unconditionally: non-attempters are coded as zero for both loan success and denial. This avoids conditioning on a post-treatment outcome, since BLSM itself affects whether households attempt to borrow. As a descriptive check, I also estimate loan success among households that attempted to borrow. In that selected sample, the success effect is small and insignificant (+0.023, SE = 0.024; Appendix Table 8). Taken together, the unconditional and conditional estimates suggest that BLSM increased entry into external borrowing rather than changing approval probabilities among applicants.

The extension remains subject to the limits of a two-period DiD design. Because the borrowing module is unavailable in IFLS3, pre-trends for borrowing cannot be tested directly. The results should therefore be interpreted as matched-DiD evidence conditional on the maintained parallel-trends assumption, rather than as definitive proof of the borrowing mechanism.

5.7 BLT I cross-sectional check

Table 9 presents cross-sectional OLS estimates on matched IFLS4 data. BLT I recipients are 3.1 percentage points less likely to have attempted borrowing (−0.031, $p = 0.018$), 2.7 points less likely to have obtained a loan (−0.027, $p = 0.039$), and show lower borrowed amounts (−0.286 log points) and outstanding balances (−0.913 log points, both $p < 0.01$). These differences are consistent with a debt-reducing association of BLT I but cannot be read causally, since IFLS3 lacks the BH module and so provides no pre-program baseline. They are offered as descriptive evidence only; the contrast with the causally identified *positive* BLSM extensive-margin effect

underscores the danger of interpreting the cross-section causally, and may reflect that the 2005 BLT and the 2013–14 BLSM operated in very different credit environments.

6 Discussion and Conclusion

This paper replicated Al Izzati, Suryadarma, and Suryahadi (2023) and extended their analysis to household external borrowing. The replication confirms the paper's core finding: short-term UCTs in Indonesia did not systematically alter behavioural outcomes or preferences across employment, smoking, informal-insurance participation, working hours, risk aversion, and intertemporal choice. Nineteen of 21 behavioural coefficients, and all four preference coefficients, match the paper in statistical significance.

The extension qualifies rather than contradicts the replication paper. Hypothesis 1—that BLSM reduces reliance on external borrowing—is rejected: recipients became *more* likely both to attempt external borrowing ($+0.045$, $p = 0.018$) and to obtain a loan ($+0.050$, $p = 0.008$). Because attempt and success move together (correlation ≈ 0.98) while the success rate among attempters is unchanged, the effect is one of *market entry*—more households participating in external credit—rather than of higher approval among existing applicants. This pattern is consistent with a financial-inclusion or creditworthiness channel (Hoffmann et al. 2024; Bastagli et al. 2016; Torkelson 2020; Bartik et al. 2024), though the IFLS does not record disbursement channel or lender type, so the mechanism is suggested rather than demonstrated.

The heterogeneity analysis points in the direction Morduch's (1999) framework predicts—a positive participation effect among less-poor recipients, with a negative (substitution-like) offset among the poorest—but the differential is not statistically significant, largely because the corrected expenditure-poverty measure classifies only about 2% of households as poor, leaving the interaction underpowered. The robust contribution is therefore the *average* increase in credit-market participation, not a demonstrated difference across the welfare distribution.

6.1 Limitations and avenues for further work

Several limitations qualify these conclusions. First, the borrowing extension relies on a two-period DiD design. Because IFLS3 lacks the BH borrowing module, I cannot estimate a comparable BLT I borrowing DiD or test pre-trends for the BLSM borrowing outcomes. The extension therefore rests on the maintained parallel-trends assumption after CEM and household fixed effects. Future work could strengthen identification by using additional administrative variation, such as district-level program coverage, as an instrument for BLSM receipt.

Second, the poverty heterogeneity results are limited mainly by statistical power. Baseline `poor` status is included in the extension CEM, so treated and control households are compared within matched cells that condition on poverty status. However, the strict expenditure-poverty line classifies only about 2% of households as poor. As a result, the triple-interaction estimate is imprecise and should be read as suggestive rather than conclusive. A broader vulnerability measure, such as a near-poor threshold or a continuous expenditure moderator, would use more of the welfare variation in the sample.

Third, some borrowing outcomes are not fully comparable across waves. In particular, IFLS4 records repayment and outstanding balance for the largest loan, while IFLS5 records these variables across all loans. I therefore interpret the intensive-margin results more cautiously than the extensive-margin outcomes.

Fourth, the matched samples may still be selective. The extension CEM retains about 67% of treated households, while the replication CEM retains a larger share of treated observations. This difference in common support may affect comparability across exercises. In addition, I do not implement a formal attrition correction, so differential panel retention remains a possible source of bias.

Finally, binary outcomes are estimated as linear probability models for consistency with Al Izzati, Suryadarma, and Suryahadi (2023). Re-estimating the borrowing outcomes with probit or logit models would be a useful robustness check to confirm that the results are not driven by predictions near the probability bounds.

6.2 Policy implications

Taken altogether, the replication supports the Al Izzati, Suryadarma, and Suryahadi (2023) conclusion that short-term UCTs do not produce systematic unintended behavioural consequences across the labor-market, consumption, and preference outcomes studied. The extension adds that, rather than reducing external borrowing, BLSM appears to have *broadened* households' engagement with formal external credit (i.e., more households entered the credit market and obtained loans).

One plausible policy implication is therefore that short-term UCTs can be deployed to mitigate shocks without fear of the classic behavioural distortions (reduced work, financed temptation goods), and that they may have a secondary effect of drawing recipient households into the external non-family/friend borrowing system. Whether this credit-market entry is welfare-improving (easier access to useful liquidity) or carries risks (greater debt exposure) is yet to be determined, and establishing that is an important question for future work.

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A Tables

Table 6: Pre-matching characteristics at baseline, by UCT recipient status

	Dataset 1 (2000, BLT I)			Dataset 2 (2007, BLSM)		
Variable	Control	Diff.	p	Control	Diff.	p
Female	0.532	0.029	0.001	0.529	0.018	0.026
Primary school	0.425	0.243	0.000	0.361	0.221	0.000
Junior secondary	0.173	−0.036	0.000	0.166	−0.007	0.295
Senior secondary	0.314	−0.189	0.000	0.332	−0.168	0.000
University	0.036	−0.030	0.000	0.068	−0.057	0.000
House ownership	0.821	0.029	0.000	0.785	0.079	0.000
Safe drinking water	0.589	−0.199	0.000	0.735	−0.250	0.000
Own toilet with septic	0.521	−0.267	0.000	0.708	−0.294	0.000
Have land for farming	0.357	0.017	0.050	0.303	0.039	0.000
Urban residence	0.507	−0.162	0.000	0.550	−0.170	0.000
Poor card	0.035	0.044	0.000	0.071	0.143	0.000
Crisis assistance 1998/99	0.012	0.014	0.000	—	—	—
Observations (control)		13,891			16,375	
Observations (treated)		4,265			4,968	
CEM strata		1,150			1,860	

Notes: The table reports baseline imbalance before CEM. "Control" is the untreated-group mean, and "Diff." is the treated-minus-control difference from a baseline regression of each covariate on UCT receipt; p-values use robust standard errors. Urban residence is from the BPS community classification (SC05); poor card is *surat miskin* (KR27A). The final row reports the number of strata generated by the subsequent CEM procedure. After CEM, multivariate L1 imbalance falls to approximately zero in both datasets.

Table 7:
Replication comparison: effect on risk aversion and intertemporal choice (original Table 5)

	Risk Set 1		Risk Set 2		Time Set 1		Time Set 2	
	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours
UCT × Post	0.002	−0.014	−0.024	−0.292	0.032	0.035	−0.038	0.036
	(0.060)	(0.039)	(0.057)	(1.016)	(0.056)	(0.037)	(0.053)	(0.043)
Match?	✓		✓		✓		✓	

Notes: BLSM (Dataset 2) only. "Paper" reproduces the $UCT \times$ year coefficients from Al Izzati, Suryadarma, and Suryahadi (2023) Table 5; "Ours" reports the replicated DiD coefficient on $UCT \times Post$, with standard errors in parentheses. A match (✓) means agreement in statistical significance at the 5% level; all four preference outcomes are insignificant in both the original and the replication. Individual fixed effects, period fixed effects, and CEM weights throughout; standard errors clustered at the individual level. Set 1 has lower stakes than Set 2. Risk Set 2 in the replication is estimated with limited within-person variation, so its standard error is large. *$p < 0.05$; **$p < 0.01$.

Table 8: Conditional success and denial among attempters (secondary)

	(1) Obtained loan (cond. on attempt)	(2) Denied (cond. on attempt)
UCT × Post	0.023	0.069
	(0.024)	(0.044)
Observations	2,163	2,163

Notes: Estimated only among households that attempted to borrow (or were at baseline), so the sample is conditioned on a post-treatment outcome and the estimates are reported as descriptive secondary evidence, not as causal effects. The insignificant conditional success effect, alongside the significant *unconditional* effect in Table 3, indicates that BLSM raised successful borrowing by increasing market entry rather than approval rates. Household and period fixed effects; CEM weights; SEs clustered at the household level. *$p < 0.05$; **$p < 0.01$.

Table 9: BLT I cross-sectional check, IFLS4 only (extension)

	(1) Attempted	(2) Obtained loan	(3) Log amount	(4) Log outstanding
BLT I (UCT)	−0.031*	−0.027*	−0.286**	−0.913**
	(0.013)	(0.013)	(0.088)	(0.186)
Observations	5,498	5,498	5,495	679

Notes: Cross-sectional OLS on matched IFLS4 (2007) households; CEM weights; robust standard errors. Because IFLS3 lacks the BH borrowing module, there is no pre-program baseline, so these estimates are descriptive and *not* causal. Log transformations use $\log(x+1)$. *$p < 0.05$; **$p < 0.01$.

B Figures

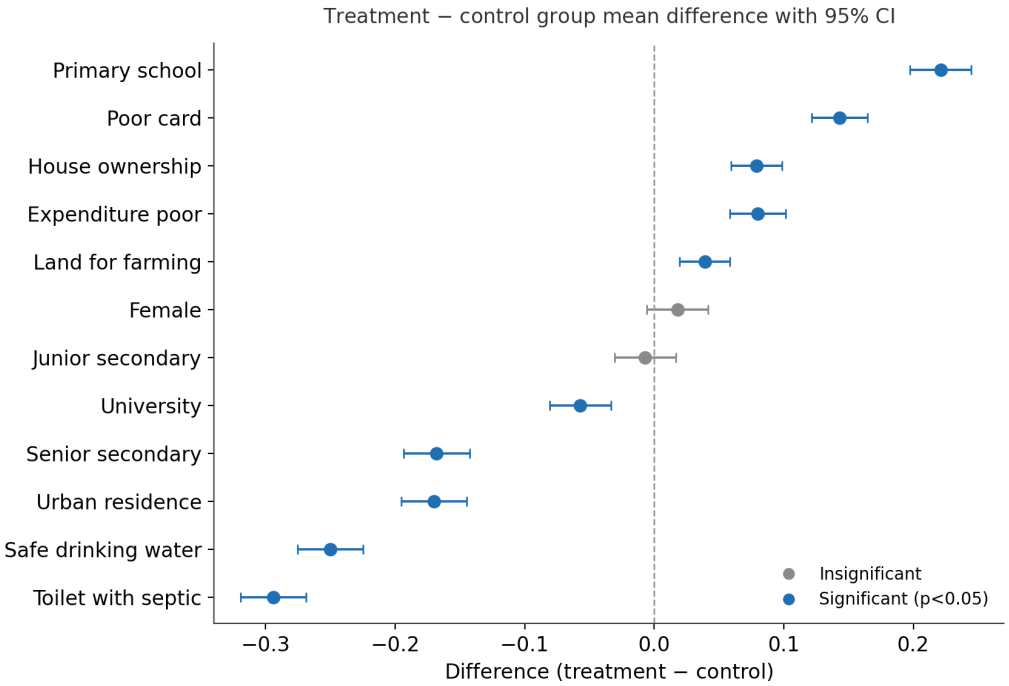


Figure 1: Pre-matching balance, Dataset 2 (BLSM, 2007 baseline). Each point is the treatment−control mean difference on a matching covariate with a 95% confidence interval; filled markers denote differences significant at the 5% level. Source: IFLS4 (2007); differences estimated by OLS regression of each covariate on UCT status.

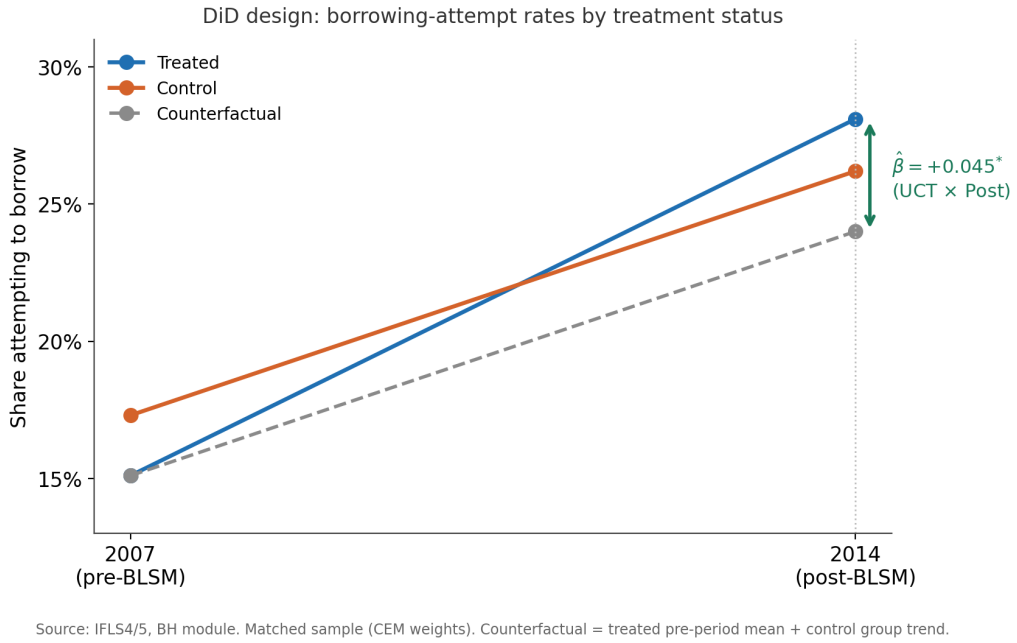


Figure 2: Difference-in-differences design for the borrowing-attempt outcome (Table 3). The dashed line is the counterfactual treated path implied by parallel trends; the bracket is the DiD estimate $\hat{\beta} = +0.045$. Source: IFLS4/5, BH module; matched sample (CEM weights).

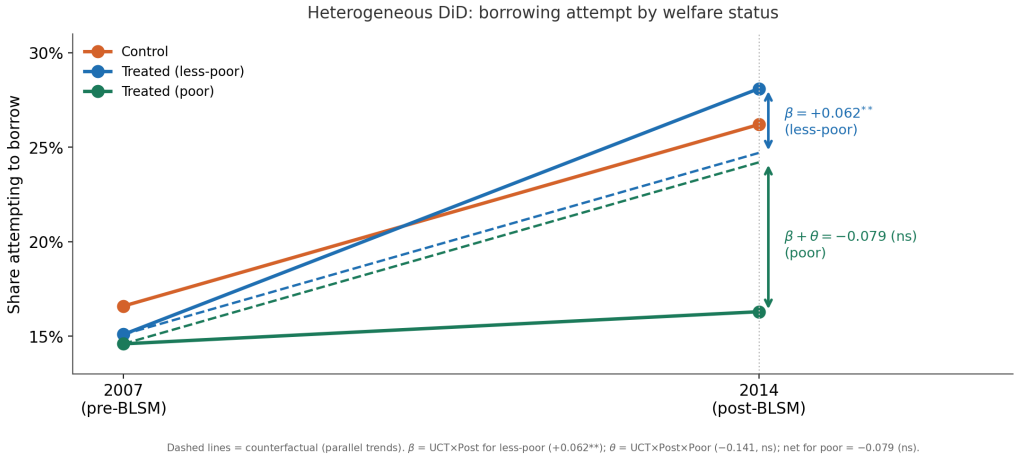


Figure 3: Heterogeneous DiD for borrowing attempt by welfare status (Table 5). Dashed lines are counterfactual (parallel-trends) paths; $\beta = +0.062$ is the less-poor effect and $\beta + \theta = -0.079$ (n.s.) is the net poor effect. The poor–non-poor difference is not statistically significant and is shown for illustration only. Source: IFLS4/5, BH module; matched sample (CEM weights).